

# Shahzaib Saqib Warraich

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*I am interested in AI interpretability and evaluation: understanding what language models truly know and building rigorous methods to measure it. My work probes internal representations, uncertainty, and predictable failure modes, bridging interpretive insight with principled benchmarking.*

## EDUCATION

**University of Southern California**  
M.S. in Applied Data Science

Jan 2024 - Dec 2025

**National University of Sciences and Technology**  
B.E. in Electrical Engineering and Computer Science

Sep 2017 - June 2021

## PUBLICATIONS

**Sample, Align, Synthesize: Graph-Based Response Synthesis with ConGrs** ([URL](#))  
S Ghosh, [SS Warraich](#), D Tarsadiya, G Yauney, S Swayamdipta ER @NeurIPS 2025

**How Reliable is Language Model Micro-Benchmarking?** ([URL](#))  
G Yauney, [SS Warraich](#), S Swayamdipta ICLR 2026 (Oral)

## RESEARCH EXPERIENCE

**Sample, Align, Synthesize: Graph-Based Response Synthesis with ConGrs** *Supervisor: Prof. Swabha Swayamdipta — ER @NeurIPS 2025*

- Developed and validated application of consensus decoding framework to factuality-critical tasks, conducting systematic evaluations across long-form biography generation and PopQA benchmarks with five independent replications per model, establishing robust baseline performance metrics and characterizing the factuality-information tradeoff
- Designed and implemented high-throughput batch processing infrastructure for FActScore evaluation, engineering performance optimizations including response caching that resolved critical processing bottlenecks and improved evaluation efficiency
- Architected comprehensive ablation studies investigating scaling dynamics of sample size during alignment and response synthesis phases, along with temperature-based sensitivity analysis, providing empirical insights into framework robustness
- Extended framework to refusal-aware settings across three challenging domains (Numerical False Presuppositions, Scientific Attributions, Historical Events), integrating sophisticated LLM-as-a-judge abstention detection for more nuanced assessment

**How Reliable is Language Model Micro-Benchmarking?** *Supervisor: Prof. Swabha Swayamdipta — ICLR 2026 (Oral)*

- Proposed and implemented confidence-based stratified sampling methodology for micro-benchmark selection in meta-evaluation frameworks, adapting principled statistical techniques from computer vision [literature](#) to language model evaluation

- Developed comprehensive evaluation infrastructure integrating multiple micro-benchmark selection approaches including stratified sampling, TinyBenchmarks (P-IRT and GP-IRT), and task reduction [baselines](#), enabling systematic comparison of benchmark selection strategies
- Maintained and extended research codebase, implementing modular components for reproducible meta-evaluation experiments across diverse benchmark selection methodologies

## INDUSTRY EXPERIENCE

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**Paramount**, Data Scientist Intern May 2025 - Oct 2025  
*Entertainment Analytics & AI*

- Enhanced content performance forecasting accuracy by up to **20%** across critical business KPIs through strategic integration of multi-platform social media engagement signals into predictive modeling frameworks
- Architected a production-grade GenAI explainability system leveraging ensemble LLM-as-a-judge methodology with majority voting consensus, enabling granular title-level performance attribution and democratizing advanced ML insights for non-technical stakeholders across the organization

**Retrocausal**, Research Engineer II (AI) Dec 2022 - Dec 2023  
*Computer Vision & Manufacturing Intelligence*

- Designed and deployed precision assembly verification solutions achieving **98%** detection accuracy for Fortune 500 manufacturing clients including Schneider Electric, Carrier, and Honda through state-of-the-art object detection and activity recognition algorithms ([Assembly Copilot](#))
- Architected an automated model training infrastructure incorporating advanced cut-paste-mix augmentation techniques to generate photorealistic synthetic datasets, reducing manual annotation overhead by **80%** while maintaining rigorous performance standards
- Conceptualized and led development of a computer vision-enabled ergonomics risk assessment module for Kaizen Copilot, establishing a differentiated product capability addressing workplace safety optimization and regulatory compliance ([Kaizen Copilot](#))

**Adlytic AI**, AI Engineer June 2021 - Dec 2022  
*Edge AI & Recommender Systems*

- Spearheaded comprehensive model optimization initiatives, implementing knowledge distillation and NVIDIA TensorRT optimization to achieve **66%** reduction in edge inference latency while maintaining performance within **1.5%** of baseline accuracy across production deployments
- Architected and operationalized a scalable cloud-based recommender system achieving **97%** prediction accuracy, driving a **3x** increase in product conversion rates and generating measurable revenue impact through enhanced customer engagement

## AWARDS & HONORS

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- USC Andrew J. Viterbi Scholarship 2024 - 2025
- William J. Fulbright Scholarship 2024 - 2025

## TEACHING

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**Course Producer**, *University of Southern California – Viterbi School of Engineering*

- CSCI 544: Applied Natural Language Processing Aug 2024 - Dec 2024
- CSCI 544: Applied Natural Language Processing Jan 2025 - May 2025

## LEADERSHIP

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**Viterbi Graduate Student Association (VGSA)**, Vice President May 2025 - Present

- Led student advocacy initiatives representing 6,000 graduate students as primary liaison between student body and university administration
- Managed strategic partnerships with internal stakeholders and external organizations to enhance graduate student resources and professional opportunities
- Delivered monthly progress reports to Viterbi School Dean, communicating student concerns and actionable recommendations

## SKILLS & INTERESTS

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### Skills

- Languages: English, French
- Programming Languages: Python, C/C++, SQL
- Deep Learning: PyTorch, Hugging Face, vLLM
- Machine Learning: Scikit-Learn, OpenCV
- Data Analysis: Pandas, Dask, PySpark
- Data Visualization: Matplotlib, Seaborn, Plotly
- Cloud: AWS Sagemaker, GCP Vertex AI
- Others: Docker, Kubernetes, MongoDB, Git, LaTeX, Figma

### Interests

- AI Ethics and Responsibility
- Hackathons
- User Experience (UX) Design